

nsEVDx: A Python Library for Stationary and Non-Stationary Extreme Value Modeling

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Summary

nsEVDx is an open-source Python package for fitting stationary and non-stationary Extreme Value Distributions (EVDs) to extreme value data. It supports non-stationary models in which the location, scale, and shape parameters can each depend on user-defined covariates. Overall, nsEVDx provides a Python-native interface for frequentist and Bayesian inference of non-stationary EVD models, enabling researchers and practitioners to quantify changing risks associated with rare and extreme events.

Statement of Need

Traditional extreme value frequency analysis assumes that the statistical properties of extreme events remain stationary over time. However, non-stationary behavior in extreme events has been documented across hydroclimatic, engineering, and financial applications. For example, in hydroclimatic systems, non-stationary behavior has been documented in flood and extreme-rainfall processes, with changes linked to factors such as urbanization and climate change (Eccles et al., 2025; Jayaweera et al., 2025; Prosdocimi et al., 2015). Such changes can alter the frequency and magnitude of extreme events over time. Accurately quantifying these evolving risks and return periods requires statistical tools that can incorporate arbitrary, time-varying covariates into all EVD parameters.

Although several existing packages support extreme value analysis, modeling non-stationary EVDs requires substantial statistical and programming expertise, particularly when Bayesian sampling algorithms are involved. nsEVDx provides an easy-to-use framework for fitting stationary and non-stationary Generalized Extreme Value (GEV) and Generalized Pareto (GPD) distributions. It makes advanced sampling methods accessible to users with varying levels of expertise in extreme value modeling while retaining the flexibility required for advanced applications.

The target audience for nsEVDx includes hydrologists, climate scientists, engineers, and risk analysts who need a reliable, flexible workflow for non-stationary extreme value modeling. By reducing the effort needed to implement both frequentist optimization and advanced Markov Chain Monte Carlo (MCMC) sampling algorithms, the software allows practitioners to focus on understanding changing risks and their implications for decision-making.

State of the Field

Extreme value distributions (EVDs), particularly the GEV and GPD, are widely used to quantify the frequency and magnitude of rare events in hydrology, climatology, engineering, and finance (Castillo, 1988; Kafle et al., 2026; Katz et al., 2002; McNeil & Frey, 2000). Increasing evidence

of non-stationarity in environmental and socioeconomic systems has motivated the development of methods that allow EVD parameters to vary with time or external covariates.

Several software packages support extreme value analysis. In the R ecosystem, `ismev` (Heffernan J. E. et al., 2003), `extRemes` (Gilleland, 2025), `climextRemes` (Paciorek, 2016), and `NSGEV` (Deville, 2026) provide tools for fitting stationary and non-stationary extreme value models. These packages provide important foundations for EVD-based analysis, although their capabilities differ in areas such as non-stationary modeling and Bayesian inference. Probabilistic programming frameworks, such as Python-based `PyMC` (Oriol Abril-Pla et al., 2023) and `NumPyro` (Phan et al., 2019), and C++-based `Stan` with interfaces like `PyStan` (Stan Development Team, 2023b) and `CmdStan` (Stan Development Team, 2023a), offer flexible tools for building custom statistical models, including extreme value models. However, these tools require users to formulate non-stationary likelihood functions explicitly for EVDs along with prior specifications, parameter constraints, and sampling configurations for each application. As a result, they may be too complex for domain-specific practitioners like hydrologists, climate analysts, or risk analysts seeking easy-to-use tools. Within Python, `pyextremes` supports extreme value analysis with features such as block maxima and peaks-over-threshold extraction, return-level estimation, frequentist inference through `SciPy`, and Bayesian inference using the affine-invariant ensemble sampler, `emcee` (Foreman-Mackey et al., 2013). However, `pyextremes` is primarily focused on stationary extreme value workflows.

These differences motivate `nsEVDx`, a flexible Python tool that provides a unified framework for both stationary and non-stationary modeling. In particular, `nsEVDx` supports user-defined covariates, parameter constraints, custom priors, and MCMC algorithms such as Metropolis-Adjusted Langevin Algorithm, MALA (Roberts & Tweedie, 1996) and Hamiltonian Monte Carlo, HMC (Betancourt, 2017; Neal, 2011) for fitting non-stationary GEV and GPD models. Compared with general-purpose probabilistic programming frameworks, `nsEVDx` provides predefined EVD likelihoods, parameterizations, and inference workflows, reducing the amount of model construction required for common extreme value analyses.

Software Design

`nsEVDx` is designed to minimize the programming burden typically associated with custom Bayesian implementations while providing a transparent and flexible framework for extreme value modeling. The overall workflow implemented in `nsEVDx` is shown in Figure 1.

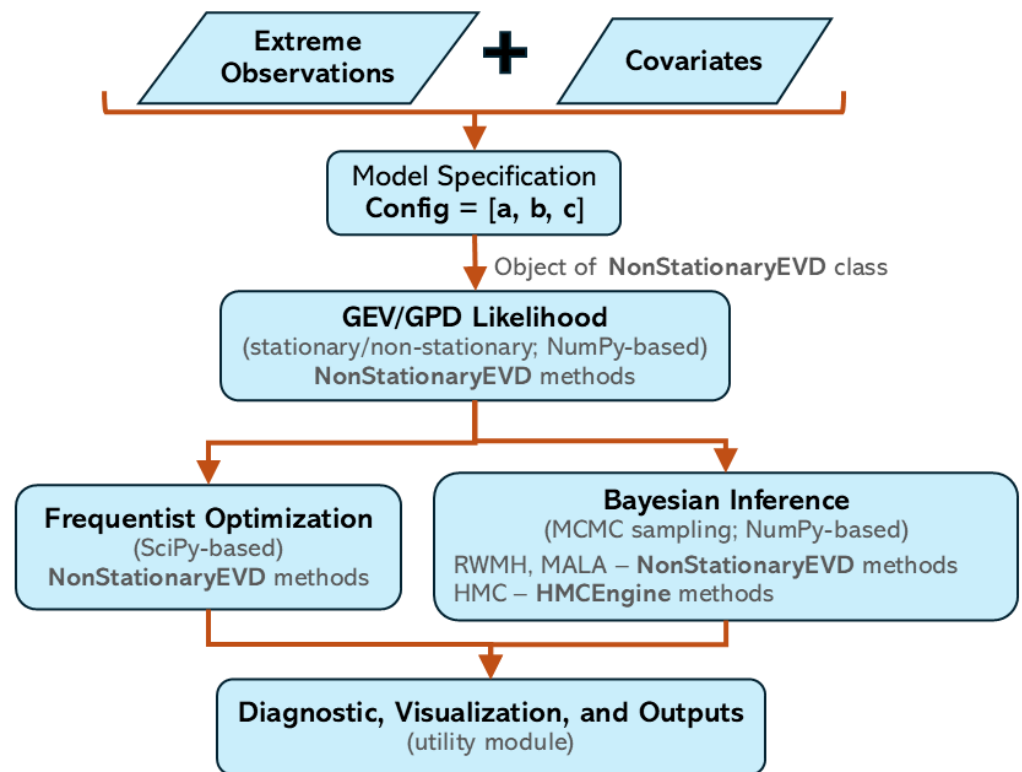
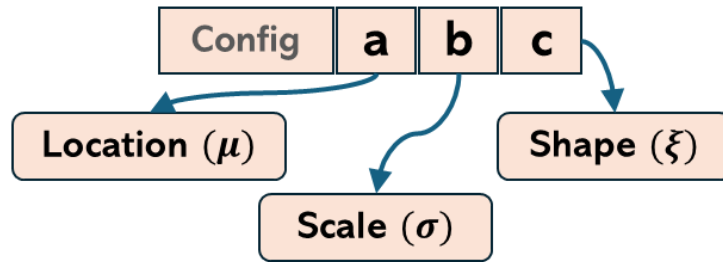


Figure 1: Workflow implemented in nsEVDx for specifying, fitting, and diagnosing stationary and non-stationary GEV and GPD models using frequentist optimization or Bayesian MCMC sampling

70 The concepts of non-stationarity and MCMC techniques used in nsEVDx are based on
 71 the foundational texts by Robert & Casella (2009) and Coles (2001). The core class
 72 NonStationaryEVD handles parameter parsing and specification, log-likelihood evaluation, prior
 73 assignment, optimization, and MCMC sampling. This unified design allows users to switch
 74 between frequentist and Bayesian workflows without redefining model structures. Frequentist
 75 methods use `scipy.optimize` to minimize the non-stationary negative log-likelihood, while
 76 the Bayesian MCMC methods are implemented in `numpy`, allowing full transparency and
 77 customization. In Bayesian estimation, nsEVDx can infer prior specifications based on
 78 the data and configuration or accept user-defined priors. In the frequentist mode, it can
 79 determine suitable parameter bounds automatically. However, user-defined priors or bounds are
 80 recommended for better convergence and interpretability. The implementation of L-moments
 81 in some utility methods follows the approach described by Hosking & Wallis (1997). Currently,
 82 nsEVDx supports linear modeling for the location and shape parameters, and exponential
 83 (log-linear) modeling for the scale parameter, to ensure positivity.

84 nsEVDx controls non-stationarity via a configuration vector `config = [a, b, c]`, where each
 85 entry specifies the number of covariates used to model the location, scale, and shape parameters
 86 of the EVD. An entry with a value of 0 implies stationarity (i.e., no covariate dependence),
 87 while integer values > 0 indicate non-stationary modeling using the corresponding number of
 88 covariates for the parameter. This representation provides a compact and scalable mechanism
 89 for defining a wide range of stationary and non-stationary models while maintaining a consistent
 90 user interface (Figure 2).



$$\mu(t) = \beta_0 + \beta_1 \cdot Z_1(t) + \beta_2 \cdot Z_2(t) + \dots + \beta_a \cdot Z_a(t)$$

$$\sigma(t) = e^{\alpha_0 + \alpha_1 \cdot Z_1(t) + \alpha_2 \cdot Z_2(t) + \dots + \alpha_c \cdot Z_b(t)}$$

$$\xi(t) = \kappa_0 + \kappa_1 \cdot Z_1(t) + \kappa_2 \cdot Z_2(t) + \dots + \kappa_c \cdot Z_c(t)$$

a , b , and c represent the number of covariates specified for modeling the location, scale, and shape parameters

$Z_1, \dots, Z_{\max(a,b,c)}$ represent the covariate vectors, and t denotes the time index
 μ , σ , and ξ represent the non-stationary location, scale and shape parameters

Examples: Config = [0,0,0] Stationary; [1,0,0] non-stationary μ ;
 [1,2,0] non-stationary μ, σ

Figure 2: Configuration-vector framework used in nsEVDx for specifying stationary and non-stationary EVD models.

91 Currently, location and shape parameters can be modeled using linear relationships, while the
 92 scale parameter is modeled using an exponential link function to ensure positivity. Polynomial
 93 relationships can be accommodated by transforming covariates before model fitting.

94 Another key design decision is the implementation of Bayesian samplers directly in NumPy. This
 95 approach maximizes transparency, enables users to inspect and modify algorithmic details, and
 96 reduces dependencies. The package currently provides Random-Walk Metropolis-Hastings
 97 (RWMH), MALA, and HMC samplers, allowing users to balance computational efficiency and
 98 inferential robustness according to their application. While RWMH and MALA are integrated
 99 within the NonStationaryEVD class, HMC is implemented through a dedicated HMC Engine
 100 component that handles Hamiltonian dynamics.

101 To support model evaluation and reproducibility, nsEVDx includes diagnostic tools for trace
 102 plots, posterior summaries, acceptance rates, and Gelman-Rubin ($\hat{R}$) convergence metrics.
 103 It also provides model selection options, including DIC, AIC, BIC, and likelihood-ratio tests,
 104 enabling complete workflows within a single environment.

105 The package is intentionally lightweight, relying primarily on NumPy and SciPy for computation
 106 and Matplotlib and Seaborn for visualization. This minimal dependency footprint improves
 107 portability and simplifies installation while preserving extensibility for future methodological
 108 developments.

109 Benchmark Experiment

110 A simulation-based case study was performed to compare mean absolute errors (MAE) in
 111 estimating EVD parameters across multiple packages, including nsEVDx. First, we generated
 112 $N = 100$ stationary and $N = 50$ non-stationary datasets, each consisting of 500 extreme values
 113 sampled using fixed EVD parameter sets (hereafter referred to as the “true” parameter; Figure
 114 3). These datasets were then used to fit EVD models, and MAE was computed by comparing
 115 estimated parameters with the corresponding true parameters. For stationary GEV and GPD

models, nsEVDx produced parameter estimates comparable to those obtained with SciPy and pyextremes using both maximum likelihood estimation (MLE) and L-moments (LM) (Figure 3a-b). For non-stationary models, nsEVDx produced parameter estimates comparable to those obtained from an equivalent NumPyro-based NUTS implementation using the same priors and MCMC settings (number of chains, posterior samples, and burn-in period), demonstrating reliable recovery of non-stationary parameters in these simulated examples (Figure 3c-d). Across methods and scenarios, shape parameters generally exhibited the lowest MAE, while differences in the MAE of other parameters were relatively small.

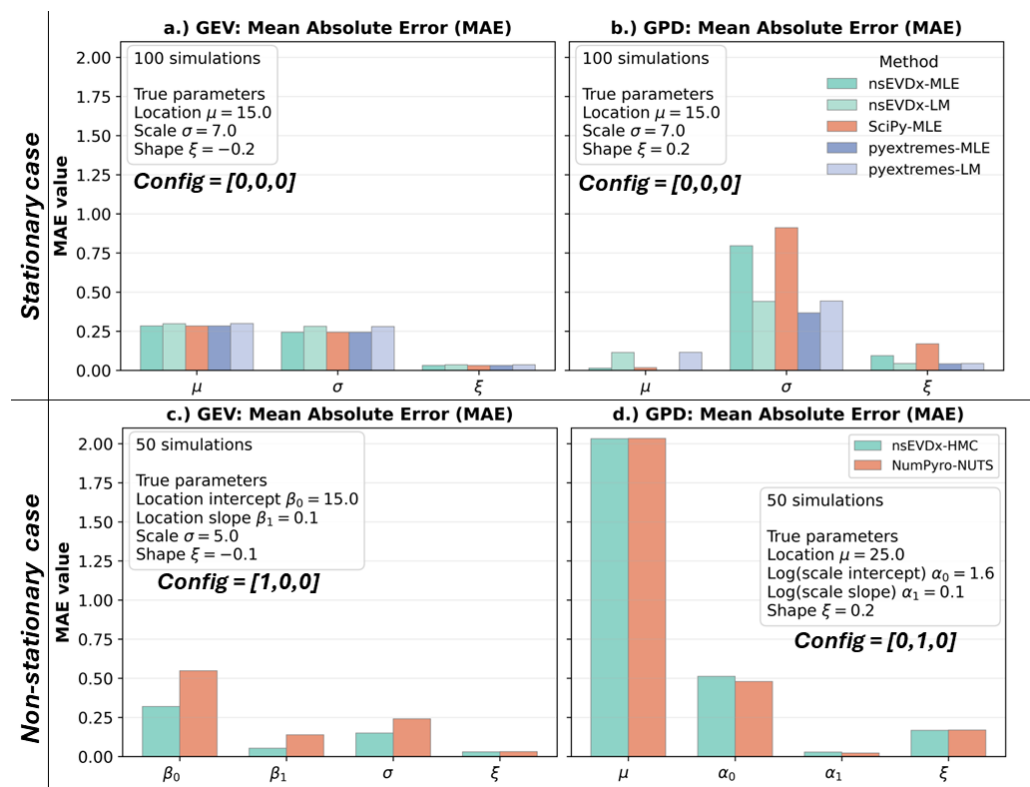


Figure 3: Mean of absolute error in estimating GEV and GPD parameters across the simulations (each using a vector of 500 random values generated using the true parameters) for various methods and Python packages.

Research Impact Statement

nsEVDx lowers the barrier to applying advanced Bayesian methods, including MALA and HMC sampling, by providing predefined extreme value models and inference workflows that accommodate users with different levels of statistical and programming expertise. The package has already been used in ongoing hydroclimatic research focused on trend detection and characterization of extreme precipitation (Kafle et al., 2025; Kafle, 2026). Potential applications extend to climate science, water resources engineering, infrastructure design, and financial risk assessment, e.g., linking rainfall extremes to temperature or maximum stock market losses to market volatility indices. A steady increase in downloads suggests growing interest among researchers and practitioners. The software is openly developed, tested, documented, and distributed through both GitHub and PyPI, supporting long-term reuse and community contributions.

Installation and Example

Install the package via pip or GitHub:

```
pip install nsEVDx
```

or

```
git clone https://github.com/Nischalcs50/nsEVDx.git
```

```
cd nsEVDx
```

```
pip install .
```

```
import nsEVDx as ns
```

```
import numpy as np
```

```
from scipy.stats import genextreme
```

```
import matplotlib.pyplot as plt
```

```
np.random.seed(123)
```

```
covariate = np.array(range(100))
```

```
config = [1,0,0] # non-stationary location; other parameters stationary
```

```
true_parameter = [30, 0.1, 5, -0.3]
```

```
Observations = ns.NonStationaryEVD.ns_EVDrvs(genextreme, true_parameter,  
                                              covariate,  
                                              config, size=100)
```

```
plt.plot(Observations)
```

```
# Inferring GEV Parameters from the Observations
```

```
model = ns.NonStationaryEVD(config, Observations,  
                           covariate,dist='GEV')
```

```
results = model.MH_Hmc(num_samples=1500, burn_in=1500,  
                      initial_params=[30, 0, 5, -0.1],  
                      num_chains=4, T = 1)
```

```
# Results
```

```
print(f"sample mean : {np.round(np.vstack(results['chains']).mean(axis=0), 3)}")
```

```
ns.plot_trace(results['chains'], config, fig_size=(8,8),show=False);
```

```
ns.plot_posterior(results['chains'], config, fig_size=(8,8),show=False);
```

Detailed installation instructions, tutorials, API documentation, and executable examples are available in the [project GitHub repository](#) and [full online documentation](#).

AI Usage Disclosure

AI tools were used to assist the development of this library and in drafting the manuscript. All underlying scientific content, software architecture, analyses, and final interpretations were conceived, verified, and approved by the authors.

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References

- Betancourt, M. (2017). A Conceptual Introduction to Hamiltonian Monte Carlo. *arXiv: Methodology*. <https://doi.org/10.48550/arXiv.1701.02434>
- Castillo, E. (1988). *Extreme value theory in engineering*. Academic Press. <https://doi.org/10.1016/C2009-0-22169-6>
- Coles, S. (2001). *An introduction to statistical modeling of extreme values* (4th. printing). Springer. <https://doi.org/10.1007/978-1-4471-3675-0>
- Deville, Y. (2026). *NSGEV: Non-stationary GEV time series*. <https://github.com/IRSN/NSGEV/>
- Eccles, R., Syktus, J., Trancoso, R., Chapman, S., Wasko, C., Evans, J. P., Thatcher, M., Di Virgilio, G., & Stassen, C. (2025). Substantial increases in future precipitation extremes—insights from a large ensemble of downscaled CMIP6 models. *Npj Natural Hazards*, 2(1), 60. <https://doi.org/10.1038/s44304-025-00107-1>
- Foreman-Mackey, D., Hogg, D. W., Lang, D., & Goodman, J. (2013). Emcee: The MCMC hammer. *PASP*, 125, 306–312. <https://doi.org/10.1086/670067>
- Gilleland, E. (2025). *extRemes: Extreme Value Analysis*. <https://doi.org/10.1175/JTECH-D-20-0070.1>
- Heffernan J. E., Stephenson A.G., & Gilleland E. (2003). *Ismev: An Introduction to Statistical Modeling of Extreme Values*. <https://doi.org/10.32614/CRAN.package.ismev>
- Hosking, J. R. M., & Wallis, J. R. (1997). *Regional Frequency Analysis: An Approach Based on L-Moments* (Vol. 93). Cambridge University Press. <https://doi.org/10.1017/CBO9780511529443>
- Jayaweera, L., Wasko, C., & Nathan, R. (2025). Evidence for non-stationarity in the GEV shape parameter when modeling extreme rainfall. *Water Resources Research*, 61(5), e2023WR036426. <https://doi.org/10.1029/2023WR036426>
- Kafle, N. (2026). *Rain-gauge network effects on the uncertainty and trends in short-duration extreme precipitation* (PhD Dissertation 32785789, The University of Memphis). <https://ezproxy.memphis.edu:3443/login?url=https://www.proquest.com/dissertations-theses/rain-gauge-network-effects-on-uncertainty-trends/docview/3369343212/se-2>
- Kafle, N., Dell'Aira, F., Chadwick, C., & Meier, C. I. (2026). Robustness of regionally derived, short-duration rainfall depth-duration-frequency estimates to the choice of minimum interevent time: Evidence across climates, raingauge densities, and regionalization approaches. *Journal of Hydrologic Engineering*. <https://doi.org/10.1061/JHYEFF.HEENG-6729>
- Kafle, N., Peleg, N., & Meier, C. I. (2025). Detecting spatially consistent trends in sub-hourly extreme rainfall using a neighborhood-based method. *AGU Fall Meeting Abstracts, 2025*, H13G–07. <https://ui.adsabs.harvard.edu/abs/2025AGUFMH13G...07K/abstract>
- Katz, R. W., Parlange, M. B., & Naveau, P. (2002). Statistics of extremes in hydrology. *Advances in Water Resources*, 25(8–12), 1287–1304. [https://doi.org/10.1016/S0309-1708\(02\)00056-8](https://doi.org/10.1016/S0309-1708(02)00056-8)
- McNeil, A. J., & Frey, R. (2000). Estimation of tail-related risk measures for heteroscedastic financial time series: An extreme value approach. *Journal of Empirical Finance*, 7(3–4),

- 271–300. [https://doi.org/10.1016/S0927-5398\(00\)00012-8](https://doi.org/10.1016/S0927-5398(00)00012-8)
- Neal, R. M. (2011). MCMC using hamiltonian dynamics. In S. Brooks, A. Gelman, G. L. Jones, & X.-L. Meng (Eds.), *Handbook of markov chain monte carlo* (pp. 113–162). CRC Press. <https://doi.org/10.1201/b10905>
- Oriol Abril-Pla, Virgile Andreani, C. Carroll, L. Y. Dong, Christopher Fannesbeck, Maxim Kochurov, Ravin Kumar, Junpeng Lao, Christian C. Luhmann, Osvaldo A. Martin, Michael Osthege, Ricardo Vieira, Thomas V. Wiecki, & Robert Zinkov. (2023). PyMC: A modern, and comprehensive probabilistic programming framework in Python. *PeerJ Computer Science*, 9, e1516–e1516. <https://doi.org/10.7717/peerj-cs.1516>
- Paciorek, C. (2016). *climextRemes: Tools for Analyzing Climate Extremes*. <https://doi.org/10.32614/CRAN.package.climextRemes>
- Phan, D., Pradhan, N., & Jankowiak, M. (2019). Composable effects for flexible and accelerated probabilistic programming in NumPyro. *arXiv Preprint arXiv:1912.11554*. <https://doi.org/10.48550/arXiv.1912.11554>
- Prosdocimi, I., Kjeldsen, T. R., & Miller, J. D. (2015). Detection and attribution of urbanization effect on flood extremes using nonstationary flood-frequency models. *Water Resources Research*, 51(6), 4244–4262. <https://doi.org/10.1002/2015WR017065>
- Robert, C. P., & Casella, G. (2009). *Introducing Monte Carlo Methods with R*. <https://doi.org/10.1007/978-1-4419-1576-4>
- Roberts, G. O., & Tweedie, R. L. (1996). Exponential Convergence of Langevin Distributions and Their Discrete Approximations. *Bernoulli*, 2(4), 341. <https://doi.org/10.2307/3318418>
- Stan Development Team. (2023a). *CmdStan: The command-line interface to Stan*. <https://doi.org/10.5281/zenodo.1117248>
- Stan Development Team. (2023b). *PyStan: The python interface to Stan*. <https://doi.org/10.5281/zenodo.1456206>